Professorship for Autonomous Aerial Systems
School of Engineering and Design
Technical University of Munich

**Eexam**

Sticker with SRID to be glued here

**Notes on personalization:**

- Your exam is personalized during the attendance check by affixing a code.
- This only contains a consecutive number, which is also noted on the attendance list next to the signature field.
- This is used as a pseudonym to enable a clear assignment of your examination.

# Engineering Mechanics 1

**Written exam:** LRG0010/ Test Exam **Examiner:** Prof. Dr.-Ing. Markus Ryll

## Instructions

- This exam consists of **26 pages** with a total of **6 assignments**. Please check now that you have received a complete set.
- It is prohibited to detach pages from the exam.
- The total score on this exam is 90 points.
- The processing time is 90 minutes.
- Permitted aids: a non-programmable calculator.
- Results must be entered in the boxes or drawings provided.
- All results are to be given in the given units and coordinates, unless otherwise requested. Formula and fraction expressions need not be simplified or summarized.
- Answer only in the provided spaces on the examination sheet. If there is not enough space, please use the additional sheets at the end. Make it clear that you have used the supplementary sheets for the respective sub-task.
- As a matter of principle, no individual content-related questions will be answered. If in your opinion, there is a supposed lack of clarity in the content of the question, this must be question in writing on the reverse side of the last page of the examination sheet. Make yourself known by a show of hands. The question will be examined. If necessary, the information will be made available to all students at the same time.
- Please do not use red color!
- Tampering with the score boxes will be penalized.

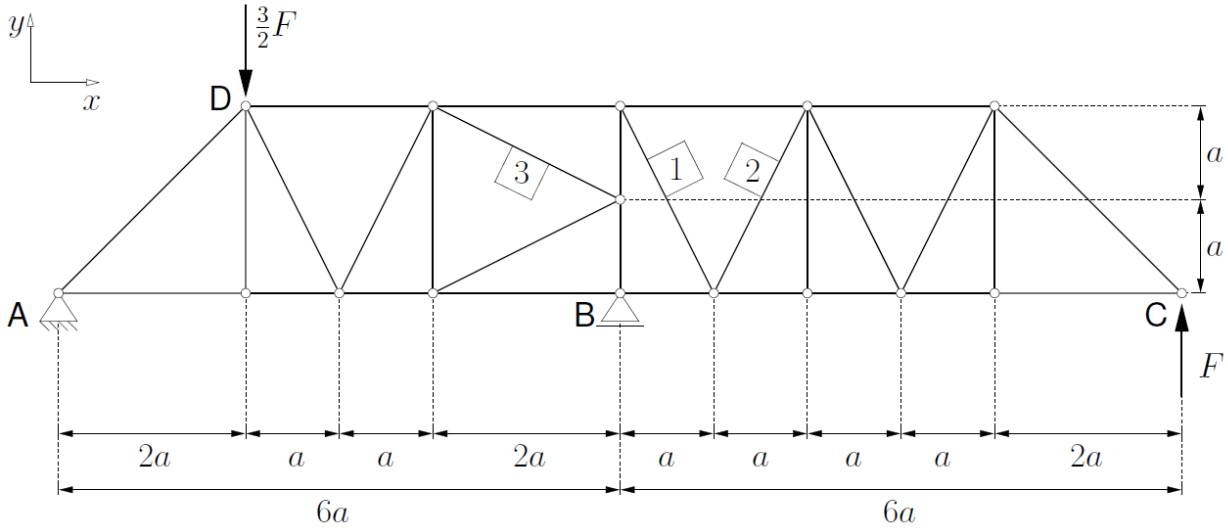
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Lecture hall left from _____ to _____ / Early drop off at _____

## **Exercise 1** (14 points)

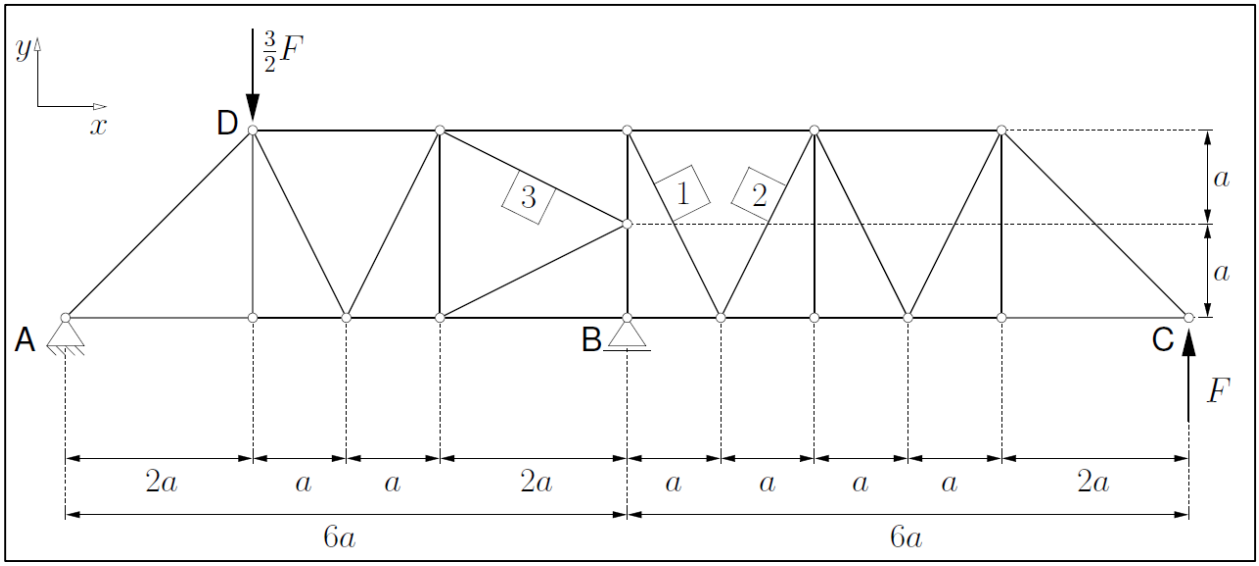
Given is the shown plane ideal truss made of rigid bars. It is rotatable at point A and movable at point B. At point C a point load $F$ is applied in positive $y$-direction. At point D, a concentrated load $\frac{3}{2}F$ is applied in the negative y-direction. The dimensions are shown in the drawing.

Given quantities: $a$, $F$, coordinate system $xy$ (right-hand system)




All subtasks can be solved independently of each other.

a) Mark clearly all zero bars of the system in the sketch. (2 Points)



b) Calculate the bar forces $S_1$ and $S_2$ in bars 1 and 2 as a result of the given load. (6 Points)

$S_1$ = $S_2$ =

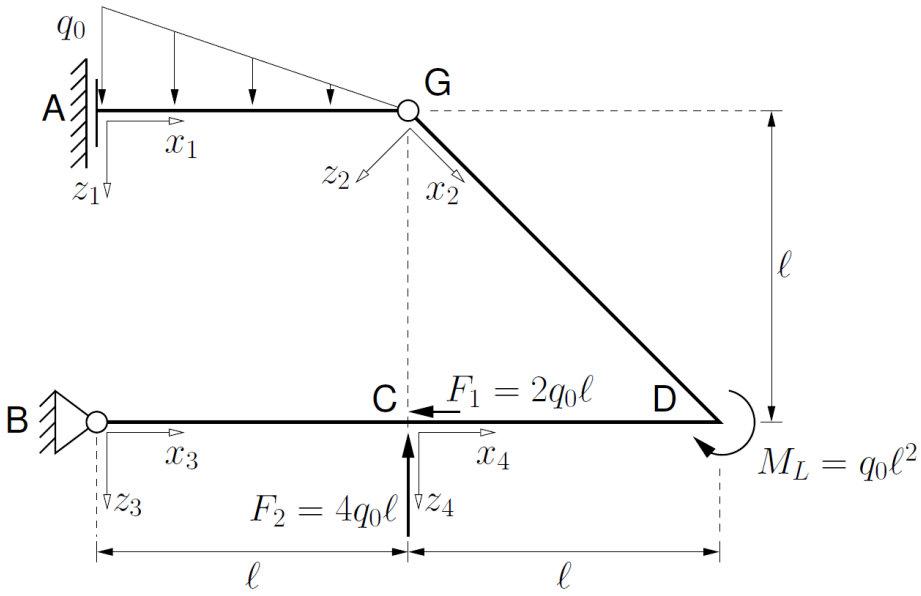
c) Calculate the bar force $S_3$ in bar $\boxed{3}$ as a result of the given load. (6 Points)

$S_3$ =

## **Exercise 2** (16 points)

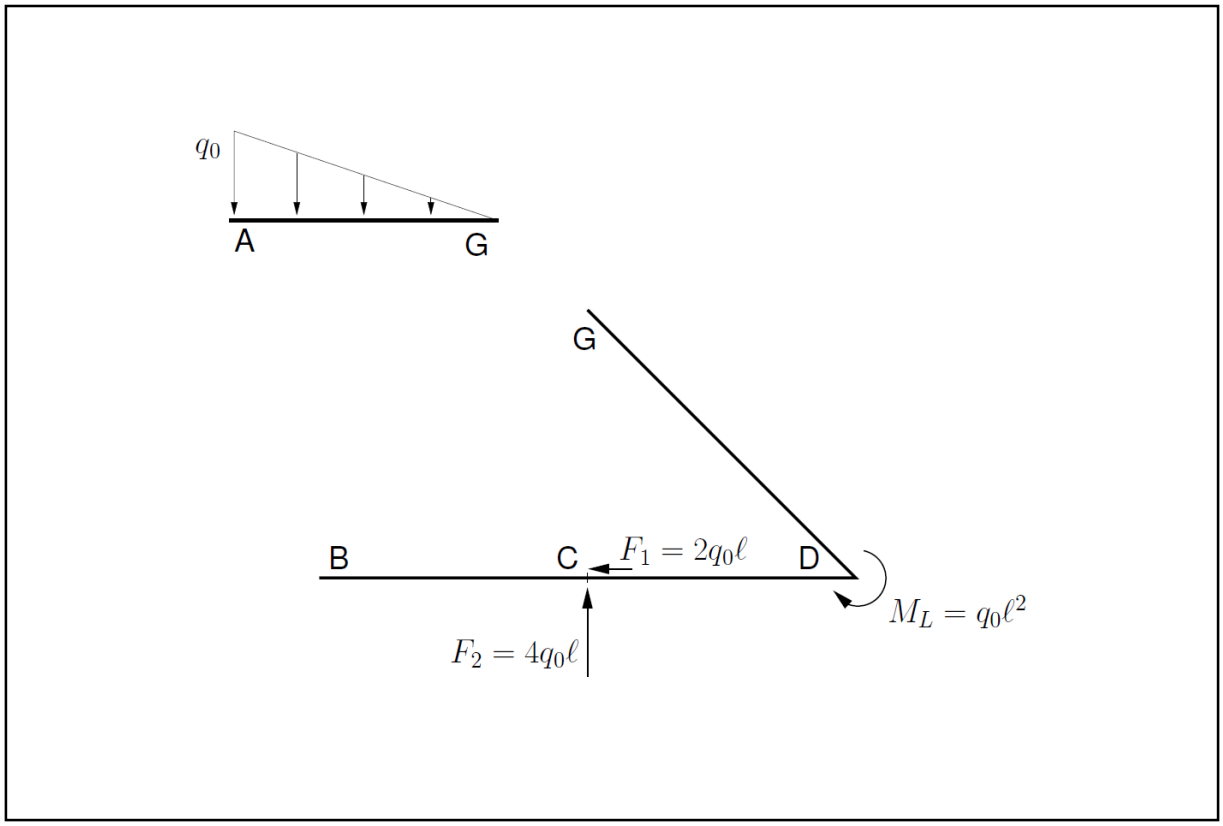
Given is the shown planar rigid frame structure, which is supported at points A and B by two supports. At point G there is a moment joint. The system is loaded by two point loads $F_1 = 2q_0 l$ and $F_2 = 4q_0 l$ at point C, which act in negative $x_4$-direction respectively negative $z_4$-direction, by a single moment $M_L = q_0 l^2$ in point D as drawn as well as in the area $x_1 \in [0, l]$ by a triangular line load in positive $z_1$-direction with maximum value $q_0$ as drawn. Bearing conditions and dimensions can be taken from the drawing.

Given quantities: $l, q_0, F_1 = 2q_0 l, F_2 = 4l, M_L = q_0 l^2$, coordinate systems $x_1z_1$, $x_2z_2$, $x_3z_3$, $x_4z_4$, (right handed systems)

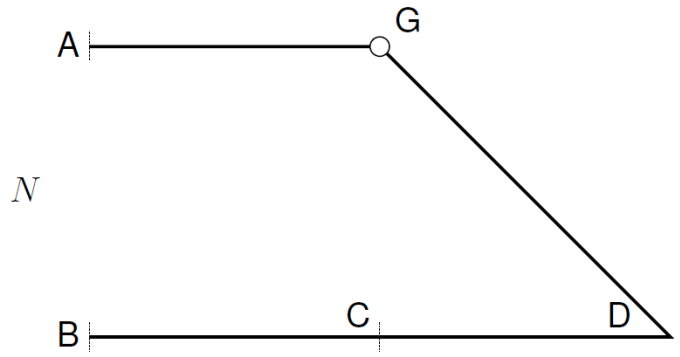


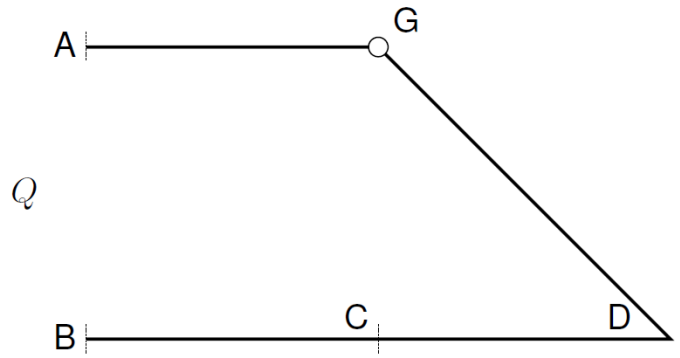

All subtasks can be solved independently of each other.

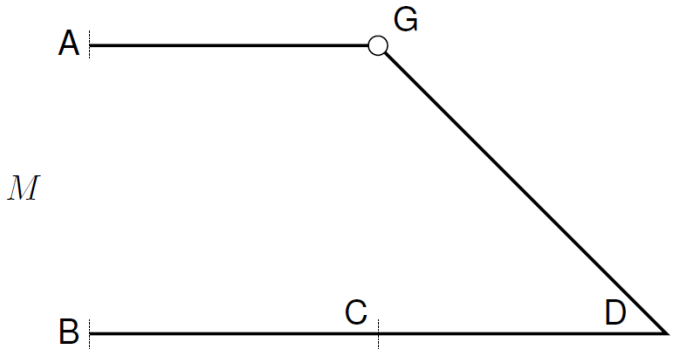
a) Draw all support and joint reactions of the system in the section cut and name them meaningfully and unambiguously. (4 Points)



b) Draw the internal forces of the normal force $N$, the shear force $Q$ and the bending moment $M$ in the entire structure in the sketches provided. Specify the sign, zeros, maxima and minima, polynomial order, characteristic values and tangents parallel to the structure. Use the coordinate systems given in the specification sketch. (12 Points)


A
G
N
B
C
D

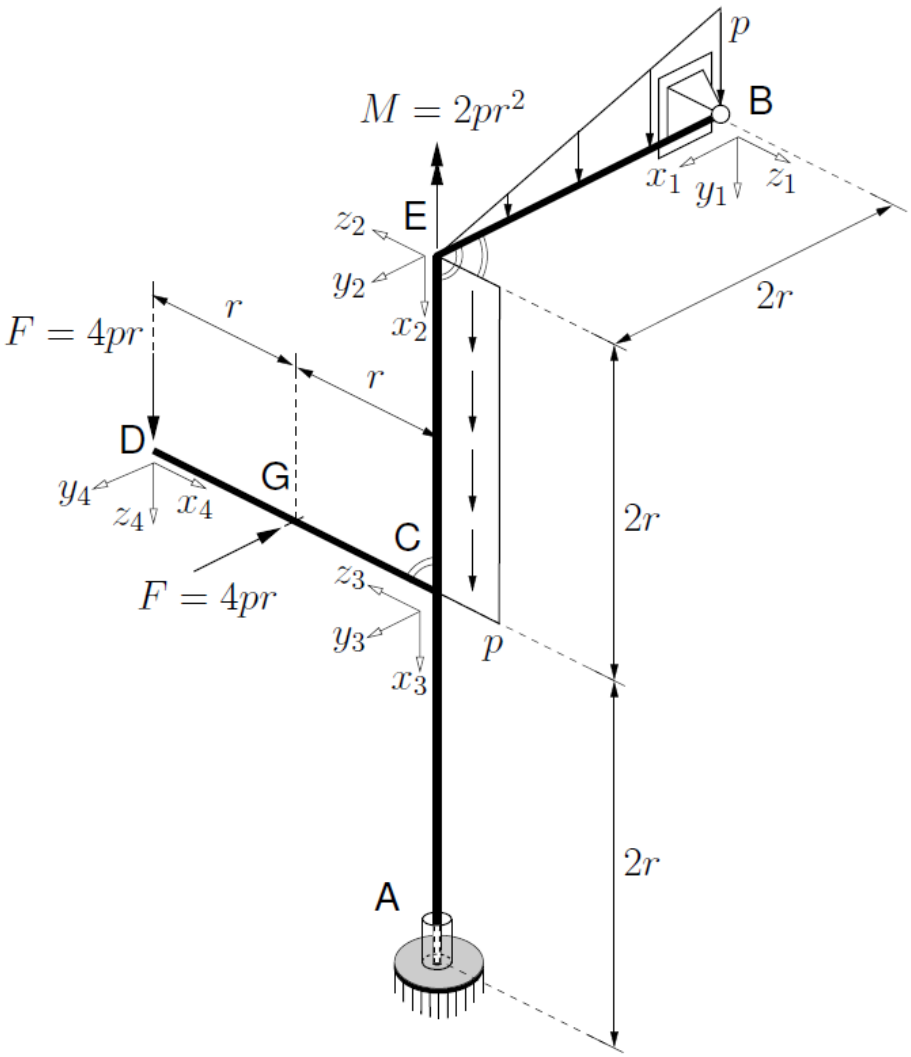

A
G
Q
B
C
D


A
G
M
B
C
D

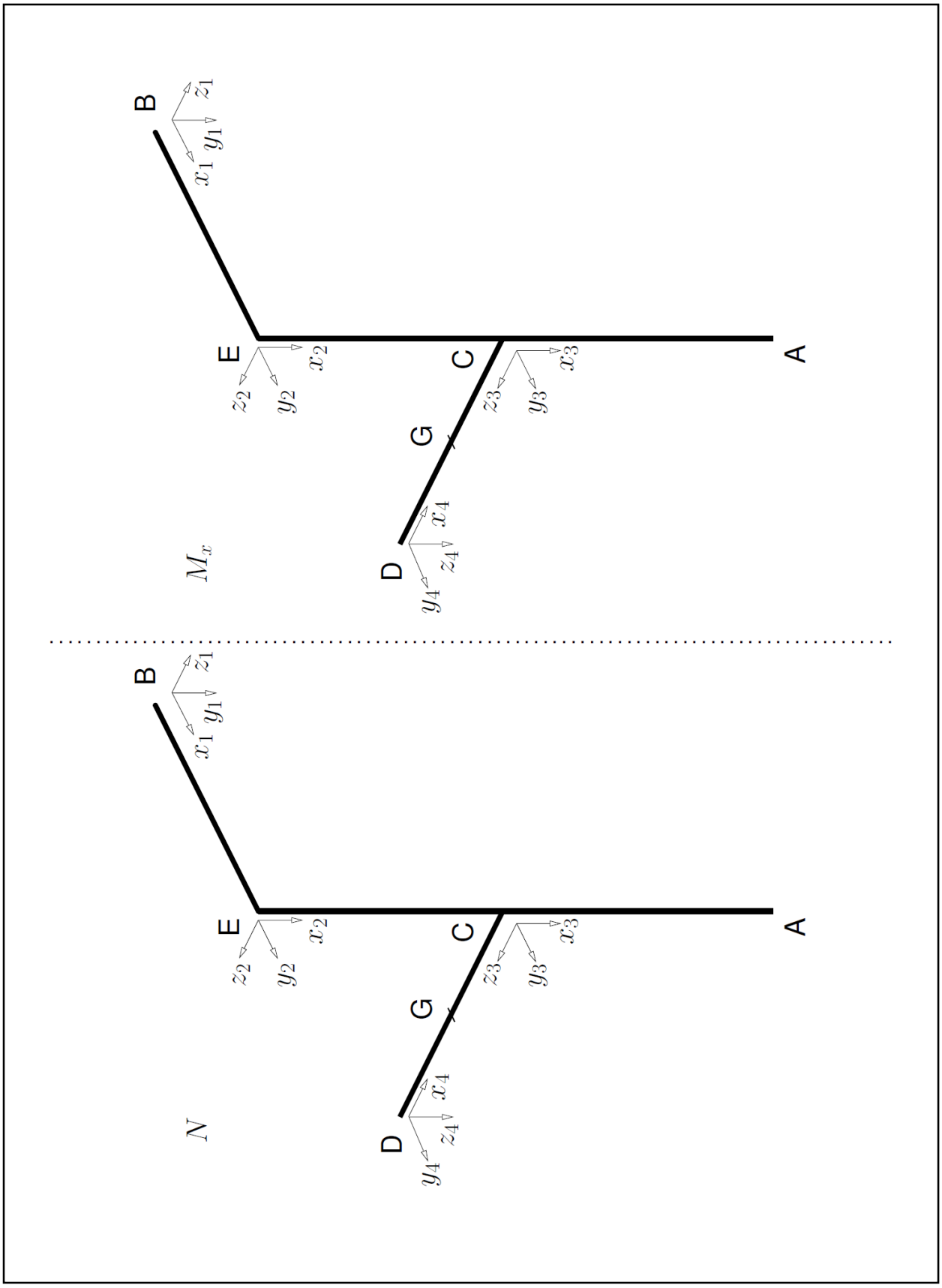
## **Exercise 3** (23 Points)

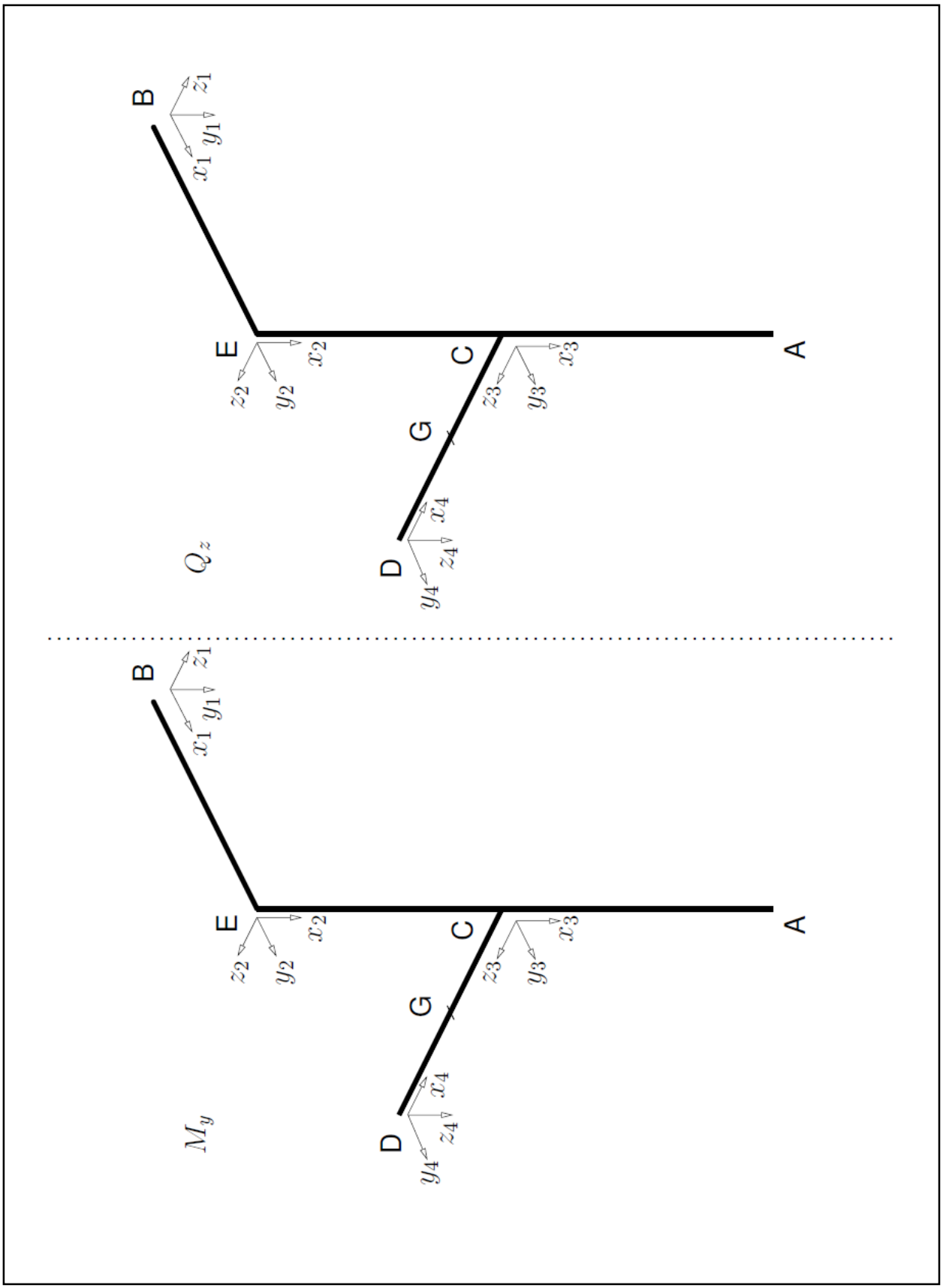
Given is a one-piece rigid frame shown below. It is supported at point B by a 1-value support and at point A by a 5-value support which only permits rotation about the $x_3$-axis. At the free end D, a point load $F = 4pr$ acts in the positive $z_4$-direction. Another concentrated load $F = 4pr$ is applied at point G in the negative $y_4$-direction. In beam section BE, a triangular distributed load with maximum value $p$ acts in positive $y_1$-direction as drawn. At point E, a single moment $M = 2pr^2$ acts in negative $x_2$-direction. In the beam section CE, a constant distributed load $p$ acts in positive $x_2$-direction. Dimensions as well as local coordinate systems can be taken from the drawing.

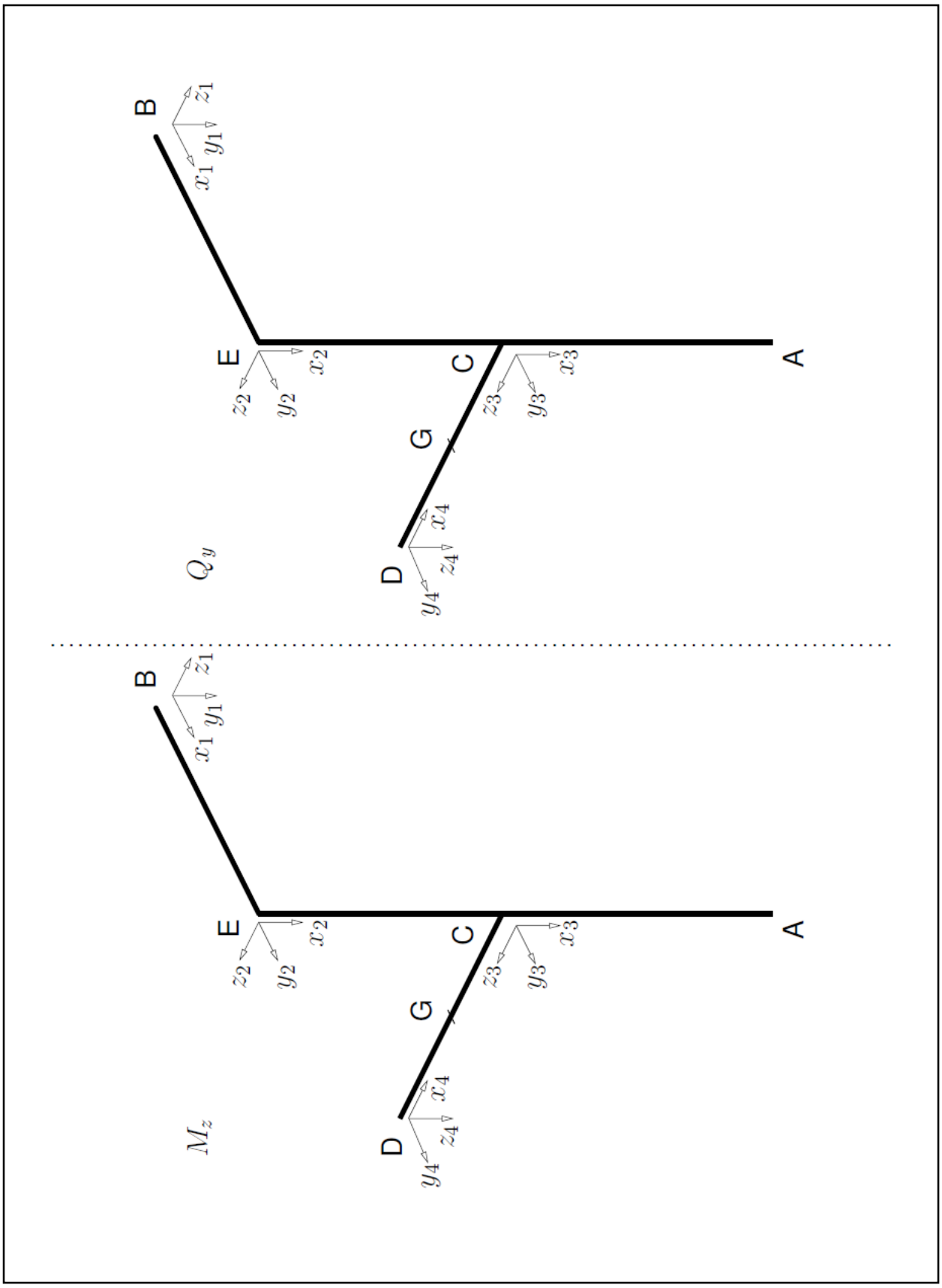
Given quantities: $r, p, F = 4pr, M = 2pr^2$, coordinate systems $x_1y_1z_1$, $x_2y_2z_2$, $x_3y_3z_3$, $x_4y_4z_4$, (right handed systems)



Draw all internal forces in the entire structure in the sketches. Specify the sign, zero points, maximum and minimum points, polynomial order, characteristic values and tangents parallel to the structure. Use the coordinate systems given in the sketch. (23 Points)


B
E
C
A
G
D
$z_1$
$y_1$
$x_1$
$x_2$
$z_2$
$y_2$
$x_3$
$z_3$
$y_3$
$x_4$
$z_4$
$y_4$
$M_x$
B
E
C
A
G
D
$z_1$
$y_1$
$x_1$
$x_2$
$z_2$
$y_2$
$x_3$
$z_3$
$y_3$
$x_4$
$z_4$
$y_4$
$N$

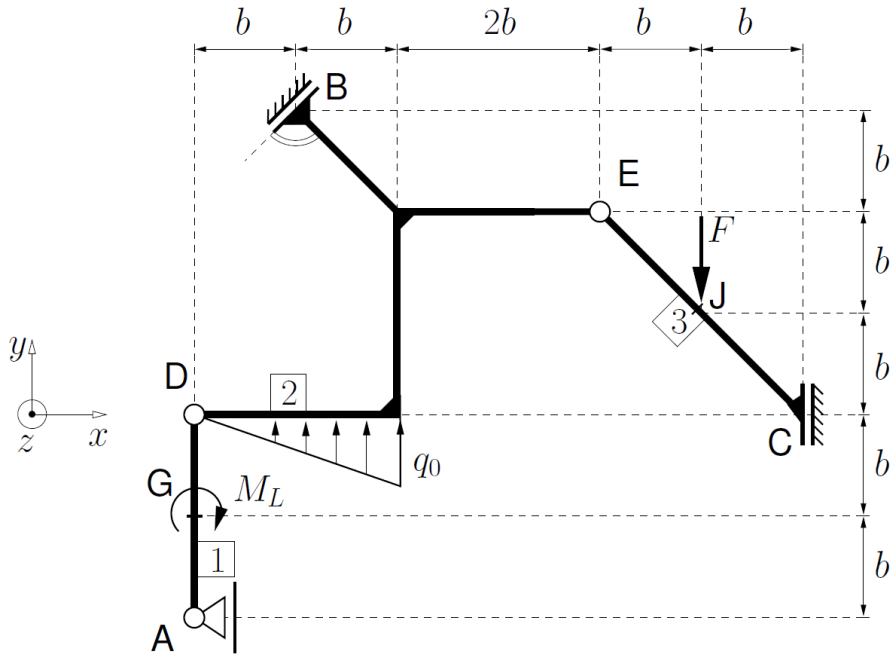

B
E
C
A
G
D
$z_1$
$y_1$
$x_1$
$x_2$
$z_2$
$y_2$
$z_3$
$y_3$
$x_3$
$x_4$
$z_4$
$y_4$
$Q_z$
$M_y$


B
E
C
A
G
D
$z_1$
$y_1$
$x_1$
$x_2$
$z_2$
$y_2$
$x_3$
$z_3$
$y_3$
$x_4$
$z_4$
$y_4$
$Q_y$
$M_z$

## **Exercise 4** (14 Points)

Given is the shown planar system of rigid frames and beams. The beam 1 is supported by a sliding support in point A. Frame 2 is hinged to the beam 1 at point D and supported by a parallel glide support at point B. Beam 3 is hinged to frame 2 at point E and is also supported with a parallel glide support at point C. The load is composed of a triangular line load with maximum value $q_0$, which acts on frame 2 in the positive y-direction as shown, a single moment $M_L$ at point G as shown, and a single load $F$ at point J, which acts in the negative y-direction. The dimensions can be taken from the drawing.

Given quantities: $b$, $q_0$, $F$, $M_L$, coordinate system $xyz$ (right-hand system)




The system is to be analyzed exclusively with the principle of virtual work. Specify the usual steps for this procedure clearly and comprehensibly.

Determine the support moment $M_{Bz}$ at point B. Assume the positive direction of action of the support moment $M_{Bz}$ is the positive z-direction. (14 Points)

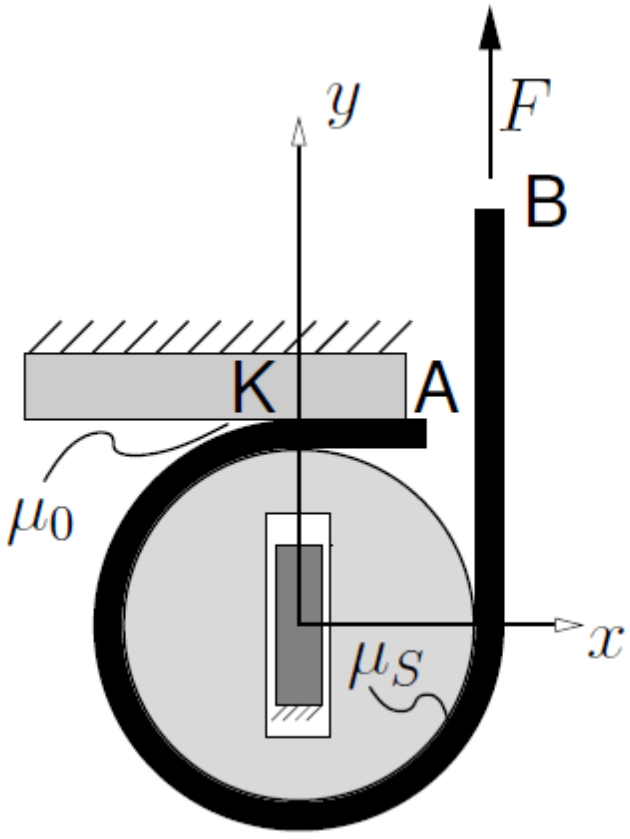
$M_{Bz}$ =

## **Exercise 5** (7 Points)

Given is the shown planar system. A rigid homogeneous circular disk is supported for displacement along the y-axis. A displacement in the x-direction and its rotation are blocked. A non-elastic rope is guided around the circular disk in a three-quarter arc. The static friction coefficient $\mu_S$ between the circular disk and the rope is known and given. At point K, the rope is also in contact with a fixed plate with an unknown coefficient of static friction $\mu_0$. At the end of the rope B, the force $F$ acts in the positive y-direction.

Note: The cross-sectional dimensions of the rope are negligibly small compared to the dimensions of the disk.

Given quantities: $\mu_S$, F, coordinate system $xy$ (right-hand system)



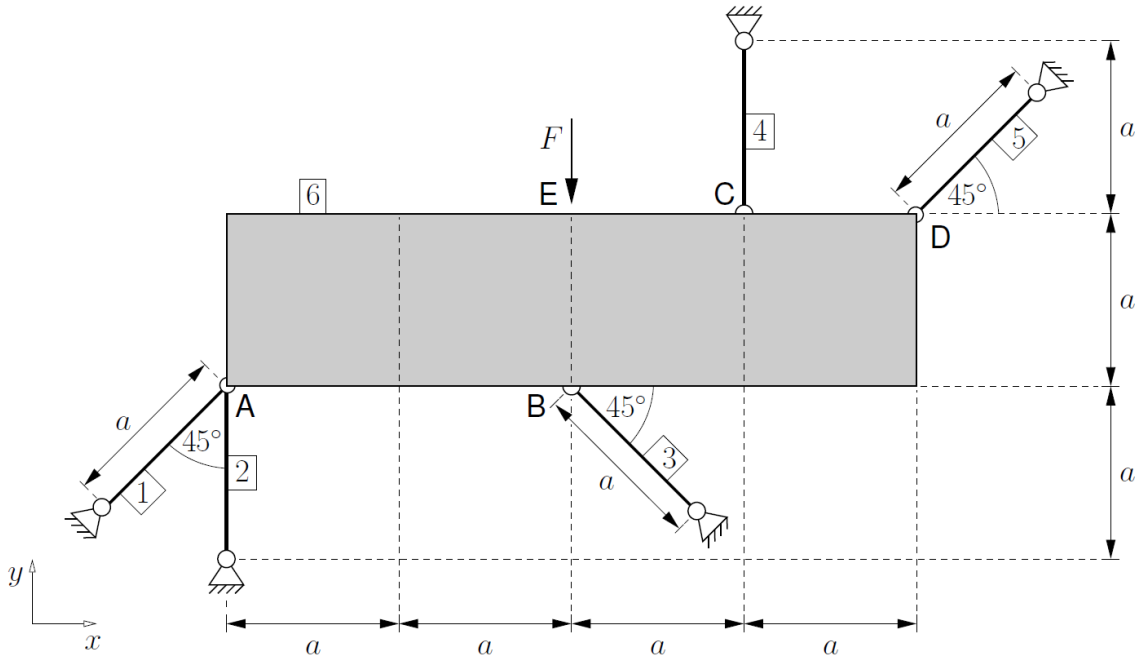
Determine the minimum required static friction coefficient $\mu_0$ on the plate as a function of $\mu_S$, so that the rope end A just does not start to slip. (7 Points)

$\mu_0$ =

**Exercise 6** (16 Points)

The shown planar system is given. The rigid rectangular disk 6 is supported at points A, B, C and D by five hinged expansion rods 1, 2, 3, 4 and 5 (each with length $a$ and constant stiffness $EA$). Thus, the system is 2-fold statically indeterminate. The load at point E is a force $F$ in the negative y-direction. Support conditions and dimensions can be taken from the drawing.

Given quantities: $a, F, EA,$ coordinate system $xy$ (right-hand system)




Solve this exercise using only the force magnitude method. Use the member force $S_4$ in rod 4 as the statically indeterminate $X_1$ and the rod force $S_5$ in rod 5 as the statically indeterminate $X_2$.

Calculate the beam force $S_4$ in beam $\boxed{4}$ due to the given load. Indicate the usual steps for this procedure clearly and comprehensibly.

$S_4$ =

**Additional space for solutions. Mark clearly and unambiguously the assignment to the respective subtask. In the original editing area, point out that parts of the solution are on the additional sheets. Do not forget to cross out invalid solutions.**